10 Radiometry and thermal radiation

10.1 Radiant energy

Electromagnetic radiation transports energy, which becomes available to heat objects, to give us information about objects it is reflected from, or to do useful work in generating electricity, for example. In this chapter we will study "radiant energy accounting"[1], to learn how much energy is available for such processes.

[1] The formal term is **radiometry**.

A useful image to work with is that radiant energy is carried by a stream of **photons**, small packets of radiant energy travelling at the speed of light. This seems very different from the model of radiation as an **electromagnetic wave**. But the two models are in fact equivalent, and there is a link between the two: The energy E carried by a photon is proportional to the frequency f of the equivalent wave. So X-ray photons have much more energy than visible light photons, and these in turn have more energy than infrared photons. This relation is called **Planck's law**, and is expressed symbolically as

In the case of visible light, we perceive different frequencies as different colours. So a red photon, for example, has less energy than a violet photon.

$$E = hf \tag{10.1}$$

where $6.62 \times 10^{-34}\,\mathrm{J\,Hz^{-1}}$ is known as **Planck's constant**.

A photon can be *emitted* by a molecule, whose energy accordingly *decreases* by the amount carried off by the photon. A photon can also be *absorbed* by a molecule, in which case the molecule's energy *increases* by the same amount. A photon can also be *reflected* by an electron. It also happens that a molecule absorbs a photon and immediately re-emits it in a random direction; in this case, the photon is said to be **scattered** by the molecule.

If we adopt the photon model, we can think of sunlight, for example, as photons streaming out from the Sun equally in all directions, with a wide range of energies, but on average corresponding to the yellow colour typical of the Sun.

Radiometry is a very jargon-rich discipline, so in the rest of this chapter we will be introducing many new, and carefully defined terms. It will help to work with some concrete scenarios: We will begin by considering how photons are used to generate electricity in solar panels.

10.2 A practical application: Photovoltaics

One source of renewable energy that is gaining a lot of traction today is **photovoltaics**. Solar panels capture radiant energy from the Sun, and convert some of it[2] to electricity.

Solar photons ionize atoms in the solar panel, and under the action of a voltage, the liberated electrons accelerate, creating a current. The power delivered by a solar panel array depends therefore on three things:

- how much energy the Sun is emitting each second;
- what fraction of that energy reaches 1 square meter of the surface of the Earth; and
- the collecting area and orientation of the solar panel array.

In this chapter, we will develop the tools necessary to answer the first two items. Then, given the power requirements, we can calculate the third. In this way, we will learn some of the fundamental concepts of **radiometry**, the study of how light is measured.

[2] About 5-20% for commercial systems, and up to 42% for the best prototype high-efficiency solar cells. The theoretical maximum efficiency is 86%.



Figure 10.1: Solar panels by Chandra Marsono.

10.3 Radiometric quantities

Radiant flux, emittance, and irradiance

Radiant energy flowing through a surface per unit time is called a **radiant flux**, Φ_e, and it is naturally measured in W. For example, the radiant flux from the Sun is an unimaginable 3.84×10^{26}W. It is emitted uniformly in all directions.

When this flux reaches a surface, like the surface of the solar panels in our example, the flux per unit area of the surface is called the **irradiance** E_e [3] , and it is measured in $\mathrm{W\,m^{-2}}$.

To work out the irradiance at the location of a solar panel on Earth, facing the Sun directly, consider an imaginary sphere centered at the Sun and with a radius equal to the distance r from the Sun to the solar panel on the surface of the Earth[4]. The flux emitted by the Sun, *which is the same in all directions*, is spread out over this sphere, so that the irradiance at any point on the sphere (and in particular, at the location of our solar panel) is

$$E_e = \frac{\Phi_e}{4\pi r^2} = 1.37\ \mathrm{kW\,m^{-2}} \tag{10.2}$$

In order to know how much power reaches the panel, we have to multiply the irradiance by the area A of the panel. Thus, for example, a $10\mathrm{m}^2$ array would receive a total flux of $E_e A = 13.7$ kW.

[3] If instead we are talking about the radiant flux per unit area being *emitted* by a source like the Sun, then it is called the **emittance** M_e, which of course is still measured in $\mathrm{W\,m^{-2}}$. We will talk more about the emittance of a source like the Sun later, when we look at thermal radiation.

[4] Also known as the Astronomical Unit (AU), which is 1.50×10^{11} m.

Generally, though, a solar panel does not face the Sun squarely, so its normal is at an angle θ with respect to the incoming rays. If the area of the panel is A, its **projected area** perpendicular to the incoming rays is only $A\cos\theta$, so the flux incident on the panel is less. For example, if our 10m^2 array was at 30° to the incoming rays, the irradiance would fall to

$$E_e = (1.37\ \text{kW m}^{-2}) \times \cos 30° = 1.18\ \text{kW m}^{-2} \tag{10.3}$$

and the flux on the solar array would be 11.8 kW.

Radiant intensity

Before going further, please read Appendix C to familiarize yourself with the concept of solid angles, and the definition of the **steradian**.

In principle, a source may radiate differently in different directions. Imagine a very narrow cone of solid angle $d\Omega$ with its vertex at the source. If the flux travelling away from the source within that cone is $d\Phi$, then the **radiant intensity** I_e is defined as $d\Phi/d\Omega$. It is measured in W sr^{-1}.

We will look at two particular instances: an **isotropic**[5] point source of radiation, and an extended source of radiation where the flux is a function of direction.

[5] "Isotropic" means the same in all directions.

Radiant intensity from an isotropic point source

In this case, the flux Φ is independent of direction, and therefore the radiant intensity is simply

$$I_e = \frac{\Phi}{4\pi} \tag{10.4}$$

To a very good approximation, an example of an isotropic point source is the Sun, because it's so far away[6]. If we compare the expression of eq. (10.4) with the irradiance for an isotropic point source of eq. (10.3), we see that

[6] The diameter of the Sun is about 1% of the distance to the Earth.

$$E_e = \frac{I_e}{r^2} \tag{10.5}$$

Since I_e is a constant in this case, the irradiance at a distance r from the Sun follows an **inverse square law**:

The radiant intensity of the Sun is $3.05 \times 10^{25}\text{W sr}^{-1}$.

$$E_e \propto \frac{1}{r^2} \tag{10.6}$$

Direction-dependent radiant intensity from an extended source – an example

In computer graphics, an interesting problem is how to render an illuminated surface realistically. Light may be **scattered** in all directions from a surface, but not necessarily equally. There are two extreme cases:

1. **Perfect specular reflection**: The surface acts like a mirror. Light incident at an angle θ with respect to the normal is entirely reflected at the same angle.
2. **Perfectly diffuse reflection**: The surface scatters the incident light equally in all directions, independently of the direction of the incident light.

In the case of diffuse reflection, the radiant intensity is equal to the incoming flux spread out over the hemisphere surrounding the surface, which has a solid angle of 2π steradians:

$$I = \frac{\Phi_e}{2\pi} \tag{10.7}$$

and the incident flux is equal to the irradiance E_e multiplied by the area A of the surface:

$$\Phi = E_e A \tag{10.8}$$

Now, as we have seen, the irradiance is proportional to $\cos\theta$. For the same incoming flux, if the direction is perpendicular to the surface so that $\theta = 0$, the irradiance is maximum. Call this E_{max}; then if the direction of the incoming light is at an angle θ to the normal,

$$E_e = E_{max} \cos\theta \tag{10.9}$$

and therefore

$$I = E_{max} A \cos\theta = I_{max} \cos\theta \tag{10.10}$$

where we have called $E_{max}A$ the maximum radiant intensity I_{max}. This is called **Lambert's law**, and a surface with this property is called a **Lambertian surface**.

Figure 10.2 shows a sphere rendered with a Lambertian surface, illuminated from the same direction that we're looking from, for simplicity. It's quite a good rendering to the extent that it does give the impression of a sphere, with a matte surface. Figure 10.3 shows the gaseous planet Jupiter illuminated in the same way, a good example of Lambertian scattering.

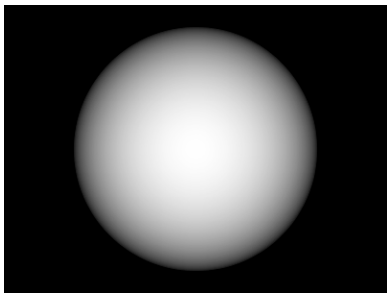


Figure 10.2: A sphere rendered with Lambertian shading



Figure 10.3: The planet Jupiter.

10.4 Thermal radiation

In general, there are three ways in which an object can interact with radiation:

- It can absorb radiation;
- It can reflect radiation; and
- It can emit radiation.

Radiation coming from outside is absorbed or reflected. But how does an object *emit* radiation?

10.5 Emittance and spectral emittance

The key radiometric quantity that we will be working with now is the emittance M_e, which we defined earlier as the radiant energy flux per unit area emitted by an object[7]

As we shall see, the amount of energy emitted by an object depends on the wavelength of the radiation: More energy is emitted at some wavelengths than others. So we define the **spectral emittance**

$$M_\lambda = \frac{dM_e}{d\lambda} \tag{10.11}$$

In this way, we recover the emittance by integrating the spectral emittance over all wavelengths:

$$M_e = \int_0^\infty \frac{dM_e}{d\lambda} d\lambda = \int_0^\infty M_\lambda d\lambda \tag{10.12}$$

[7] The "unit area" in the definition of emittance refers to the area of the radiating object.

10.6 Blackbody radiation

One mechanism that is responsible for emission from absolutely all objects is simply due to the fact that they have a non-zero temperature. We consider first an idealized object called a **blackbody**, characterized by the property that if any radiation falls on it, it is entirely absorbed, and none is reflected. In other words, any radiation that we see coming from the object was generated by the object itself, and is not merely reflected from some other light source.

Many objects behave as blackbodies to a very good approximation. Two useful instances to have in mind as we proceed are:

- Stars, including of course our Sun; and
- incandescent bulb filaments.

These are useful examples because they are very different in everything except their radiant properties, and because we find them in a wide range of temperatures.

A blackbody in thermal equilibrium contains not only atoms, but also photons. We can imagine the blackbody as a container at a temperature T, of volume V, enclosing a "gas" of photons. Just like an ordinary gas in a box, the photons don't all have the same energy: Some have more energy and some have less. We say that the photon gas has an **energy distribution**. Since by Planck's law the energy of a photon corresponds to a value of the wavelength, it follows that we can express the energy distribution in terms of wavelength.

Let's call the energy per unit volume, or energy density, of the blackbody u. Because this energy is carried by photons of different wavelengths, it is convenient to define a **spectral energy density** u_λ in the same way that we defined the spectral emittance earlier:

$$u_\lambda = \frac{du}{d\lambda} \tag{10.13}$$

so that the energy density may be found by integrating the spectral energy density over all wavelengths:

$$u = \int_0^\infty \frac{du}{d\lambda} d\lambda = \int_0^\infty u_\lambda d\lambda \tag{10.14}$$

The spectral energy density of the photon gas in a blackbody was derived by Max Planck, and it has the following rather forbidding expression:

$$u_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} \tag{10.15}$$

where h is Planck's constant, c is the speed of light in vacuum, and k is a universal constant known as Boltzmann's constant, which has a value of $1.38 \times 10^{-23} \mathrm{J\,K^{-1}}$, so that the combination kT is in fact an energy.

Before we investigate this function in more detail, it is important to emphasize that *all blackbodies have the same spectral energy density*. It doesn't matter how big they are, and it doesn't matter what they are made of. The only parameter that matters is the temperature.

10.7 *Exploring the blackbody spectrum*

Imagine now that the blackbody container leaks photons to the outside. In fact, this is how the Sun or a light bulb are visible. The photons that are emitted by the blackbody have the same energy distribution as the photon gas inside.

NEVER, EVER, LOOK AT THE SUN, EVEN OUT OF THE CORNER OF YOUR EYE! You will lose your eyesight permanently almost immediately.

Now remember that the function u_λ in equation (10.15) is the *energy per unit volume* contained in the photon gas inside the blackbody, per

unit wavelength. Let's calculate how much energy leaves the blackbody each second, again per unit wavelength.

Let's pick a blackbody with a simple shape: A spherical star will do nicely. We can also assume that the radiation is isotropic. In a time δt, all the photons contained in a layer of depth $c\delta t$ will emerge from the star[8]. The volume of that layer is $dV = 4\pi R^2 c\delta t$, where R is the radius of the star. So to get the amount of energy per unit wavelength leaking out of the star during δt we have to multiply the spectral energy density by that volume, that is

$$u_\lambda 4\pi R^2 c\delta t \tag{10.16}$$

and to get the amount of energy that leaves the star each second per unit wavelength, we just have to divide by δt. This is the energy flux per unit wavelength carried by photons of wavelength λ,

$$\Phi_\lambda = u_\lambda 4\pi R^2 c \tag{10.17}$$

Now $4\pi R^2$ is just the surface area of the star, so the flux per unit wavelength per unit area, i.e. the *spectral emittance* of photons of wavelength λ is

$$M_\lambda = u_\lambda c = \frac{8\pi hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT} - 1} \tag{10.18}$$

There are several important observations that we can make from the expressions for M_λ and Φ_λ:

1. M_λ depends only on the temperature T, and it is exactly the same for every blackbody. Bearing in mind that the only distinguishing feature of a blackbody is that it does not reflect any radiation, this is quite a remarkable finding.

2. If we integrate the expression for M_λ over all wavelengths, we get the total emittance from a blackbody. It turns out to be proportional to T^4. Merging all the constants, the emittance is simply

   $$M_e = \sigma T^4 \tag{10.19}$$

   where $\sigma = 5.67 \times 10^{-8} \mathrm{W\,m^{-2}\,K^{-4}}$ is called the **Stefan-Boltzmann constant**. This means that the emittance rises very sharply with temperature: If the temperature of a blackbody is doubled, the emittance rises by a factor 16!

3. It is time to plot M_λ to see what the function actually looks like. Figure 10.4 shows the spectral emittance as a function of wavelength for three different temperatures. In general, the function has a peak and drops off to zero as $\lambda \to 0$ or $\lambda \to \infty$.

   Beyond the general shape, it is evident that the area under the curve increases sharply with temperature. The area under the curve

[8] Think of this as the outermost, transparent layer of the star. Inside the star, photons are absorbed by atoms and re-emitted in random directions all the time. In the outermost layer, however, the density of the gas is so low that photons reaching it from inside the star will not be appreciably absorbed, and therefore pass straight through and into space.

Equation (10.19) is known as the **Stefan-Boltzmann law**.

is the integral of M_λ over all wavelengths, which as we saw earlier is the emittance, which is proportional to T^4 (eq. (10.19)).

It is also evident that the peak shifts towards shorter wavelengths as the temperature increases. The range highlighted in yellow in the figure is the visible range, approximately 0.4–0.7 microns[9]. For the 5000 K curve, approximately 30% of the energy emitted by the blackbody lies within the visible range, so we would see the object glow; but for the 3000 K curve, only 9% lies within the visible range, so we would hardly see it, as most of the energy would be in infrared wavelengths and longer.

[9] 1 micron is 1µm, or 10^{-6} m.

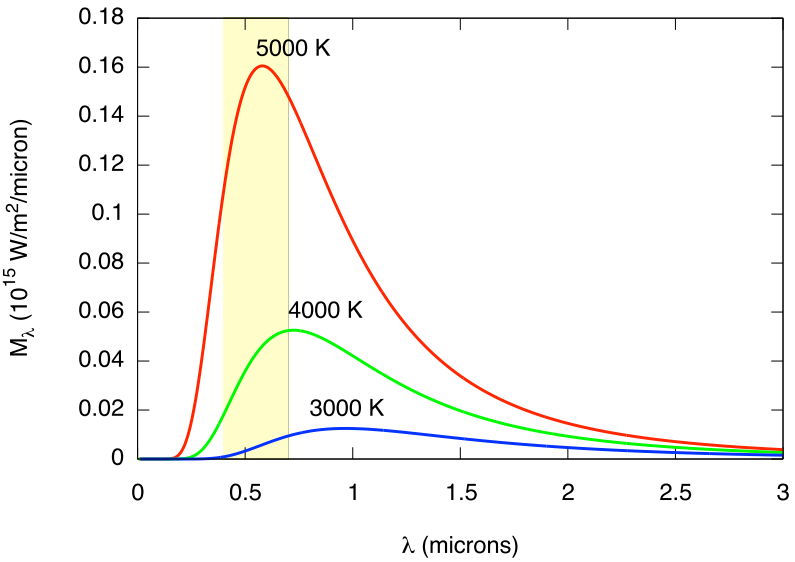


Figure 10.4: Plots of $M_\lambda(\lambda)$ at three different temperatures. The highlighted region represents the visible range of wavelengths.

10.8 Blackbodies in thermal equilibrium

In the previous section, we modelled a blackbody as a container filled with a photon gas, leaking photons to the outside. The energy distribution of those photons is a function of the blackbody's temperature. But shouldn't the temperature go down as the escaping photons carry off energy from the blackbody?

Yes, it should, and this is indeed a cooling mechanism by radiation. The only way a blackbody can maintain a steady temperature if it's radiating away energy is if it has an input of energy at the same rate.

A good example of this is the Sun (or any star, for that matter). As we saw at the beginning of this chapter, the Sun emits radiant energy at a rate of 3.84×10^{26}W. Since it has maintained that flux for billions of years, it must be generating energy in its core at the same rate. The

Sun's core is a nuclear fusion reactor, converting H nuclei to He nuclei, and generating photons in the process.

Another way for a blackbody to maintain thermal equilibrium is to absorb radiation from outside. In the examples that follow, we will look at how planets maintain a steady temperature by absorbing energy from the Sun at the same rate as they radiate away energy into space. But planets do not conform to the definition of a blackbody because they reflect part of any radiation that falls on them. We can still apply everything we have learned about blackbodies if we take into account only the fraction of the radiation that is actually absorbed by the planet. A planet is an instance of a **greybody** rather than a blackbody.

10.9 Greybodies

To describe a greybody, we use two coefficients: the **absorptance** a is the fraction of the incident radiation that is absorbed by the body, and the **reflectance** r is the fraction that is reflected. a and r are both numbers ≤ 1, and conservation of energy dictates that

$$a + r = 1 \tag{10.20}$$

Astronomers call the reflectance of a body its **albedo**, a Latin term meaning "whiteness", which is confusingly labelled a.

So if the irradiance on a greybody is E, the flux per square meter that is actually absorbed is aE. This also applies to the flux that is radiated away by a greybody: If we call the blackbody emittance M_{bb}, the greybody emittance is only aM_{bb}. It seems strange to calculate the emittance using the *absorptance* a, so in that case we call it the **emissivity** ε instead, and write the greybody emittance as

$$M_e = \varepsilon M_{bb} \tag{10.21}$$

10.10 Summary of radiometric quantities

Throughout this chapter we have introduced a zoo of radiometric quantities, so it is prudent to summarize them, to see clearly the differences and similarities between them:

Name	Symbol	Definition	SI units
Radiant flux	Φ_e	Radiant energy flowing through a surface per unit time	W
Irradiance	E_e	Radiant flux *incident* on a surface, per unit area	$\mathrm{W\,m^{-2}}$
Emittance	M_e	Radiant flux *emitted* by a surface, per unit area	$\mathrm{W\,m^{-2}}$
Radiant intensity	I_e	Flux per unit solid angle $d\Phi_e/d\lambda$	$\mathrm{W\,sr^{-1}}$
Spectral emittance	M_λ	Emittance per unit wavelength $dM_e/d\lambda$	$\mathrm{W\,m^{-3}}$
Blackbody emittance	M_{bb}	Emittance by a perfect blackbody	$\mathrm{W\,m^{-2}}$

Table 10.1: Radiometric quantities.